

Tangent Planes and Linear Approximations

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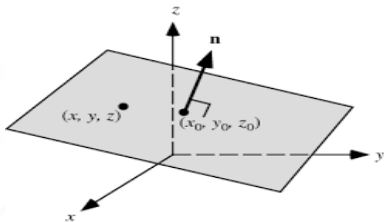
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Let us begin this section By remembering some facts ABOUT planes and lines in R^3 .

Planes The equation of a plane passing through a point $P(x_0, y_0, z_0)$ with the normal vector $n = \langle a, b, c \rangle$ can be written as

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0.$$

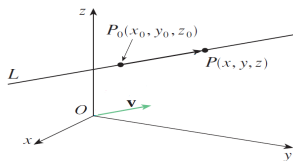


Proof

In $\mathbb{R}^3$, a straight line L in space is determined By:

- A point on L
- The direction of L

Consider the straight line below passing through a point $P_0(x_0, y_0, z_0)$ in the direction of a vector $v = \langle a, b, c \rangle$.



Proposition The equation of L can be written in the following two forms.

▶ $x = x_0 + at, y = y_0 + bt, z = z_0 + ct$
(Parametric Equation of L)

▶ $\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$
(Symmetric Equations L)

Consider a smooth surface S and a point $P(x_0, y_0, z_0)$ on it. Let

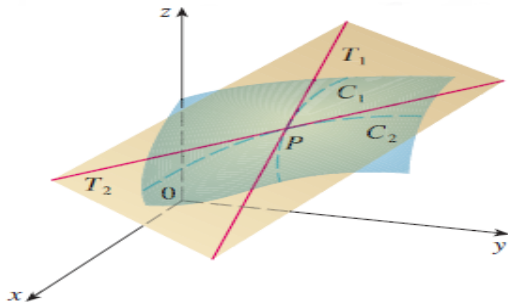
C_1 = intersection of S with the plane $y = y_0$

C_2 = intersection of S with the plane $x = x_0$

T_1 = tangent line to C_1 at the point P

T_2 = tangent line to C_2 at the point P

The tangent plane to S at P is a plane that contains both T_1 and T_2



Proposition Suppose that $z = f(x, y)$ has continuous partial derivatives. An equation of the tangent plane to the surface $z = f(x, y)$ at a point $P(x_0, y_0, z_0)$ is given By

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

The following fact is **very useful** in finding equations of tangent planes.

Let S be a surface given by the equation $F(x, y, z) = 0$. If $P(a, b, c)$ is a point on S , then the vector

$$\vec{n} = \langle F_x(P), F_y(P), F_z(P) \rangle$$

is orthogonal to S at the point $P(a, b, c)$.

Example 1. Find an equation of the tangent plane to

○ $z = e^{x-y}$ at the point $(2, 2, 1)$

$$f_x(x, y) = e^{x-y} \implies f_x(2, 2) = 1, \quad f_y(x, y) = -e^{x-y} \implies f_y(2, 2) = -1$$

and $f(2, 2) = e^0 = 1$. Using $z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$, we get

$$z - 1 = (x - 2) - (y - 2) \implies z = x - y + 1.$$

○ $z = \frac{x}{y^2}$ at the point $(-4, 2, -1)$

$$f_x(x, y) = \frac{1}{y^2} \implies f_x(-4, 2) = \frac{1}{4}, \quad f_y(x, y) = \frac{-2x}{y^3} \implies f_y(-4, 2) = -1$$

and $f(-4, 2) = -1$. Using $z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$, we get

$$z + 1 = \frac{1}{4}(x + 4) + y - 2 \implies z = \frac{x}{4} + y - 2.$$

Example 2 Find an equation of the tangent plane to the surface given by the equation

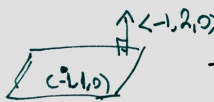
○ $x = y^2 + z^2 - 2$ at the point $A(-1, 1, 0)$

Solution

$$x = y^2 + z^2 - 2 \Rightarrow \underbrace{-x + y^2 + z^2 - 2 = 0}_{F(x,y,z)}$$

$$\Rightarrow \nabla F = \langle -1, 2y, 2z \rangle \Rightarrow$$

$$\nabla F(-1, 1, 0) = \langle -1, 2, 0 \rangle \rightarrow \text{normal vector}$$


$$\rightarrow \langle x+1, y-1, z \rangle \cdot \langle -1, 2, 0 \rangle = 0$$
$$\Rightarrow -x - 1 + 2y - 2 = 0 \Rightarrow -x + 2y = 3$$

Example 3. Consider the curve C that results from the intersection of the surface $4x^2 + 2y + z^2 = 16$ and the plane $y = 2$.

- Find the parametric equation of the tangent line to C at the point $(1, 2, \sqrt{8})$
- Find the equations of the normal line and the tangent plane to the surface $4x^2 + 2y + z^2 = 16$ at $(0, 6, 2)$.

Solution

a) $4x^2 + 2y + z^2 = 16$, $y = 2 \rightarrow$
 $4x^2 + 4 + z^2 = 16 \rightarrow 4x^2 + z^2 = 12 \Rightarrow$ we can
 parametrize C via
 $x = \sqrt{3} \cos t$, $y = 2$, $z = \sqrt{12} \sin t$.

Let t_0 be a value for which C passes through
 the point $(1, 2, \sqrt{8})$

$\Rightarrow \sqrt{3} \cos t_0 = 1 \Rightarrow \cos t_0 = \frac{1}{\sqrt{3}}$



$r(t) = \langle \sqrt{3} \cos t, 2, \sqrt{12} \sin t \rangle \Rightarrow r'(t) = \langle -\sqrt{3} \sin t, 0, \sqrt{12} \cos t \rangle$

$\Rightarrow r'(t_0) = \langle -\sqrt{3} \sin t_0, 0, \sqrt{12} \cos t_0 \rangle = \langle -\sqrt{2}, 0, 2 \rangle$

l: $\begin{cases} x = 1 - \sqrt{2}t \\ y = 2 \\ z = \sqrt{8} + 2t \end{cases}, t \in \mathbb{R} \rightarrow$ tangent line to C

An alternative solution

$$F(x,y,z) = 4x^2 + 2y + z^2 - 16 = 0 \Rightarrow \nabla F = \langle 8x, 2, 2z \rangle$$

and a normal vector to the plane $y=2$ ($\parallel$ to the xz -plane)

$$\text{is } \langle 0, 1, 0 \rangle \Rightarrow$$

$$\langle 8x, 2, 2z \rangle \times \langle 0, 1, 0 \rangle = \begin{vmatrix} i & j & k \\ 8x & 2 & 2z \\ 0 & 1 & 0 \end{vmatrix} = \langle -2z, 0, 8x \rangle$$

At the point $(1, 2, \sqrt{8})$, we get $\langle -2\sqrt{8}, 0, 8 \rangle$

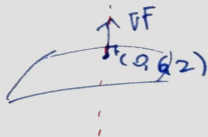
tangent $\swarrow$
to C

$$\Rightarrow l: \begin{cases} x = 1 - 2\sqrt{8}t \\ y = 2 \\ z = \sqrt{8} + 8t \end{cases}, t \in \mathbb{R}$$

is the tangent line to C .

$$\underbrace{4x^2 + 2y + z^2}_{F} = 16 \Rightarrow \nabla F = \langle 8x, 2, 2z \rangle$$

$$\nabla F(0, 6, 2) = \langle 0, 2, 4 \rangle$$



Normal line: $\begin{aligned} x &= 0 \\ y &= 6 + 2t \\ z &= 2 + 4t \end{aligned}, t \in \mathbb{R}$

Tangent Plane: $\langle x, y-6, z-2 \rangle \cdot \langle 0, 2, 4 \rangle = 0 \Rightarrow$
 $2y - 12 + 4z - 8 = 0$

Recall that the linear approximation to a function $y = f(x)$ is

$$L(x) = f(a) + f'(a)(x - a)$$

$L(x)$: An equation of the tangent line to $y = f(x)$ at $x = a$.

Linear Approximations

If $z = f(x, y)$ is a smooth function, then the **linear approximation** (or the **tangent plane approximation**) at the point $(a, b, f(a, b))$ is given by

$$L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

Remark 1. $L(x, y)$ is the equation of the tangent plane at the point $P(a, b, f(a, b))$.

Definition If $z = f(x, y)$, then f is called **differentiable** if

$$\Delta z = f_x(a, b)\Delta x + f_y(a, b)\Delta y + \epsilon_1\Delta x + \epsilon_2\Delta y$$

where

$$\epsilon_1, \epsilon_2 \rightarrow 0 \text{ as } \Delta x \rightarrow 0 \text{ and } \Delta y \rightarrow 0$$

Therefore, if $f(x, y)$ is differentiable then

$$(x, y) \sim (a, b) \implies f(x, y) \sim L(x, y)$$

There is an alternative definition of differentiability which is somewhat easier to understand.

Definition If a function $f(x, y)$ is differentiable at a point, then

- f is continuous at (a, b)

- $$\lim_{(x,y) \rightarrow (a,b)} \frac{f(x, y) - L(x, y)}{\sqrt{(x - a)^2 + (y - b)^2}} = 0.$$

Therefore, if f is differentiable at $(a, b) \implies$ the partial derivatives at (a, b) must exist.

The following theorem is sufficient but not necessary for f to be differentiable at a point.

Theorem If the partial derivatives f_x and f_y exist at and around (a, b) then f is differentiable at (a, b) .

Example. Show that $f(x, y) = \sqrt{x^2 + y^2}$ is not differentiable at $(0, 0)$.

Solution.

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{h^2}}{h} = \lim_{h \rightarrow 0} \frac{|h|}{h} = DNE$$

Similarly, $f_y(0, 0)$ does not exist either. Therefore, f cannot be differentiable at $(0, 0)$.

Example. Show that for the function

$$f(x, y) = \begin{cases} \frac{7x^2y}{x^2+y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

Both $f_x(0, 0)$ and $f_y(0, 0)$ exist, yet f is not differentiable at $(0, 0)$.

Solution.

Note that f is continuous at $(0, 0)$. Now,

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0}{h} = 0$$

Similarly,

$$f_y(0, 0) = \lim_{h \rightarrow 0} \frac{f(0, h) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0}{h} = 0$$

Thus, we have $L(x, y) = f(0, 0) + f_x(0, 0)(x - 0) + f_y(0, 0)(y - 0) = 0$.

Now, we need to check if $\lim_{(x, y) \rightarrow (a, b)} \frac{f(x, y) - L(x, y)}{\sqrt{(x - a)^2 + (y - b)^2}} = 0$ is satisfied or not.

$$\begin{aligned}
 \lim_{(x,y) \rightarrow (0,0)} \frac{f(x,y) - L(x,y)}{\sqrt{x^2 + y^2}} &= \lim_{(x,y) \rightarrow (0,0)} \frac{\frac{7x^2y}{x^2+y^2}}{\sqrt{x^2 + y^2}} \\
 &= \lim_{(x,y) \rightarrow (0,0)} \frac{7x^2y}{(x^2 + y^2)^{3/2}}
 \end{aligned}$$

Along the path $y = mx$, we obtain

$$\lim_{x \rightarrow 0} \frac{7x^2(mx)}{(x^2(1+m^2))^{3/2}} = \lim_{x \rightarrow 0} \frac{7x^3m}{x^3((1+m^2))^{3/2}} = \frac{7m}{((1+m^2))^{3/2}}$$

Therefore, the limit DNE. Hence, f is not differentiable at $(0,0)$.

Example. Show that for the function

$$f(x, y) = \begin{cases} \frac{x^2 y^2}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

is differentiable at $(0, 0)$.

Solution.

Note that f is continuous at $(0, 0)$.

As in the previous example, $f_x(0, 0) = f_y(0, 0) = f(0, 0) = 0$. Thus, $L(x, y) = 0$.

Now, we need to check if $\lim_{(x,y) \rightarrow (a,b)} \frac{f(x, y) - L(x, y)}{\sqrt{(x-a)^2 + (y-b)^2}} = 0$ is satisfied or not.

$$\lim_{(x,y) \rightarrow (0,0)} \frac{f(x, y) - L(x, y)}{\sqrt{x^2 + y^2}} = \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{(x^2 + y^2)^{3/2}} \quad (*)$$

$$0 \leq \frac{x^2 y^2}{(x^2 + y^2)^{3/2}} \leq \frac{x^2}{x^2 + y^2} \frac{\sqrt{y^2}}{\sqrt{x^2 + y^2}} \sqrt{y^2} \leq |y|$$

Thus, by the squeeze theorem, $(*)$ is zero.

Remark The red and the blue fractions above are both ≤ 1 .

Example 2 Is the given function differentiable at the indicated point? Find the linearization at that point.

1. $f(x, y) = \sqrt{x + e^{4y}}$ at the point $(3, 0)$

2. $f(x, y) = \sin(2x + 3y)$ at the point $(-3, 2)$

Solution.

$$\begin{aligned} 1. \quad f_x &= \frac{1}{2} (x + e^{4y})^{-1/2} \Rightarrow f_x(3, 0) = \frac{1}{2} 4^{-1/2} = \frac{1}{4} \\ f_y &= \frac{1}{2} (x + e^{4y})^{-1/2} \cdot 4e^{4y} \quad f_y(3, 0) = \frac{1}{2} 4^{-1/2} \cdot 4 = 1 \end{aligned} \quad \left. \begin{array}{l} f(3, 0) = 2 \\ \Rightarrow \end{array} \right\} \begin{aligned} L(x, y) &= 2 + \frac{1}{4}(x - 3) + 1 \cdot (y - 0) \\ L(x, y) &= \frac{x}{4} + y + \frac{5}{4} \end{aligned}$$

2. $z = f(x, y) = \sin(2x + 3y)$ at $(-3, 2)$

$$\left. \begin{aligned} f_x &= 2 \cos(2x + 3y) \\ f_y &= 3 \cos(2x + 3y) \end{aligned} \right\} \begin{aligned} f_x(-3, 2) &= 2 \\ f_y(-3, 2) &= 3 \end{aligned} \quad -f(-3, 2) = 0$$

$$\Rightarrow \begin{aligned} L(x, y) &= 0 + 2(x + 3) + 3(y - 2) \\ &\quad \text{so} \end{aligned}$$

$$L(x, y) = 2x + 3y$$

Example 3. Suppose that f is a differentiable function such that

$$f(2, 5) = 6, \quad f_x(2, 5) = 1, \quad f_y(2, 5) = -1.$$

Use linear approximations to estimate $f(2.2, 4.9)$.

Solution

$$f(x, y) \approx L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

$$x = 2.2, \quad a = 2$$

$$y = 4.9, \quad b = 5$$

$$\begin{aligned} f(2.2, 4.9) &\approx L(2.2, 4.9) = f(2, 5) + f_x(2, 5)(2.2 - 2) + f_y(2, 5)(4.9 - 5) \\ &= 6 + 1 \cdot 0.2 - (-0.1) \end{aligned}$$

$$= 6.3$$

Example 4. Find the linearization to $f(x, y) = \sqrt{x^2 + y^2}$ at the point $(3, 4)$. Use this linearization to estimate $f(3.04, 3.98)$. What is the **error** in this estimation?

Solution.

$$f_x = \frac{x}{\sqrt{x^2 + y^2}} \quad , \quad f_y = \frac{y}{\sqrt{x^2 + y^2}}$$

$$L(x, y) = 5 + \frac{3}{5}(x-3) + \frac{4}{5}(y-4)$$

$$f(3.04, 3.98) \approx L(3.04, 3.98) = 5.008$$

But

$$f(3.04, 3.98) = 5.00819$$

$$\begin{aligned} \text{Error} &= \text{Real Value} - \text{Estimate Value} = 5.00819 - 5.008 \\ &= 0.00019 \quad (\text{underestimate}) \end{aligned}$$

Definition If $z = f(x, y)$ then we define the differential (or total differential) of z to be

$$dz = f_x dx + f_y dy = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$$

Similarly, if $w = f(x, y, z)$ then

$$dw = f_x dx + f_y dy + f_z dz$$

Example 5. Find the differentials of

○ $u = x^2 y^3 - x + 3y$

$$du = 2xy^3 dx + 3x^2 y^2 dy - dx + 3 dy = \frac{1 \cdot (x+2y) - x}{(x+2y)^2} dx - \frac{2x}{(x+2y)^2} dy = \frac{2y}{(x+2y)^2} dx - \frac{2x}{(x+2y)^2} dy$$

○ $w = \frac{7y-3x}{z}$

Example 6. The dimensions of a closed rectangular box are measured as 80 cm 60 cm and 50 cm, respectively, with a possible error of 0.2 cm in each dimension. Use differentials to estimate the maximum error in calculating the surface area of the box.

Solution.

$$S = 2(xy + xz + yz)$$

$$dS = 2(y+z)dx + 2(x+z)dy + 2(x+y)dz$$

The max. error is $\Delta x = \Delta y = \Delta z = 0.2$, using $\Delta x = dx, \Delta y = dy, \Delta z = dz$

The max error in calculated surface area to be

$$dS = 220(0.2) + 260(0.2) + 280(0.2) = 152$$

Example 7. Use differentials to estimate the value of $3.(2.1)^2 - 2.(2.1)(-3.05) + (-3.05)^2$

Solution Let $f(x, y) = 3x^2 - 2xy + y^2$.

$$x_0 = 2, y_0 = -3, \Delta x = 0.1, \Delta y = -0.05$$

$$f(2, -3) = 33, f_x = 6x - 2y \rightarrow f_x(2, -3) = 18$$

$$f_y = -2x + 2y, f_y(2, -3) = -10$$

$$\begin{aligned} dz &= f_x(2, -3) dx + f_y(2, -3) dy \\ &= 18 \times (0.1) - 10(-0.05) = 2.3 \end{aligned}$$

$$dz \approx \Delta z = f(2.1, -3.05) - f(2, -3)$$

$$\begin{aligned} f(2.1, -3.05) - 33 &\sim 2.3 \Rightarrow \\ f(2.1, -3.05) &\sim 35.3 \end{aligned}$$

Example 8. Use differentials to estimate the maximum error in the calculated volume of a closed cylindrical can that is 10 cm high and 4 cm in diameter if the metal in the top and bottom is 0.1 cm thick and the metal in the sides is 0.05 cm thick.

Solution

③

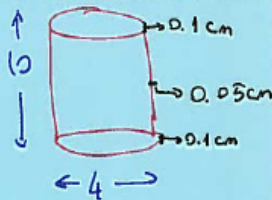
$$V = \pi r^2 h \Rightarrow$$

$$\Delta V \approx dV = 2\pi r h dr + \pi r^2 dh$$

$$dr = 0.05, \quad dh = 0.2$$

$\Rightarrow$

$$dV = 2\pi \cdot 2 \cdot 10 \cdot 0.05 + \pi \cdot 2^2 \cdot 0.2 = 2.8\pi \approx 8.8 \text{ cm}^3$$



Example 9. Consider $z = f(x, y) = x^2 + 3xy - y^2$

1. Find dz
2. If x changes from 2 to 2.05 and y changes from 3 to 2.96 find Δz and dz
3. Compare Δz and dz

Solution

$$a) \quad dz = z_x dx + z_y dy = (2x + 3y) dx + (3x - 2y) dy$$

$$b) \quad x: 2 \nearrow 2.05, \quad y: 3 \searrow 2.96, \quad \Delta z = ?, \quad dz = ?$$

$$\Rightarrow \Delta x = dx = 0.05 \quad \text{and} \quad \Delta y = dy = -0.04$$

$\Rightarrow$

$$dz = (2 \times 2 + 3 \times 3)(0.05) + (3 \times 2 - 2 \times 3)(-0.04) = 0.65$$

On the other hand,

$$\Delta z = \overset{\text{increment of } z}{f(2.05, 2.96)} - f(2, 3) = 0.6449$$

$$c) \quad dz \approx \Delta z$$

Example 10. Find the linear approximation to the function $f(x, y) = \sqrt{20 - x^2 - 7y^2}$ at $(2, 1)$. Use it to approximate $f(1.95, 1.08)$. What is the error in this estimation?

Solution

$$f_x = \frac{-x}{\sqrt{20 - x^2 - 7y^2}} \quad \text{and} \quad f_y = \frac{-7y}{\sqrt{20 - x^2 - 7y^2}}, \quad f_x(2, 1) = -\frac{2}{3}, \quad f_y(2, 1) = -\frac{7}{3}$$
$$f(2, 1) = 3$$

So, $L(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$ with $(x_0, y_0) = (2, 1)$

we have

$$L(x, y) = f(2, 1) + f_x(2, 1)(x - 2) + f_y(2, 1)(y - 1) = 3 - \frac{2}{3}(x - 2) - \frac{7}{3}(y - 1) = -\frac{2}{3}x - \frac{7}{3}y + \frac{20}{3}$$

$$f(1.95, 1.08) \sim L(1.95, 1.08) = 2.84\bar{6}$$

On the other hand,

$$f(1.95, 1.08) = 2.8342$$

$$\text{Error} = \begin{matrix} \text{Estimate} \\ \text{Value} \end{matrix} - \begin{matrix} \text{Actual} \\ \text{Value} \end{matrix} = 2.84\bar{6} - 2.8342 = 0.0124, \quad \text{it is an overestimate}$$